

Architecture Quality Checklist

Assignment 01 – Cloud Architecture Design

Before submitting your assignment, verify that your work satisfies the following quality checks. Also submit hardcopy of this checklist.

1. Problem Understanding

- ✗ The system purpose is clearly explained.
- ✗ The target users and system environment are identified.
- ✗ Functional requirements describe what the system must do.
- ✗ Non-functional requirements include scalability, security, and availability.
- ✗ Each requirement is connected to an architectural implication.

Quick self-check:

If someone reads only Section 1, can they clearly understand what system is being designed?

2. Delivery Model Design

- ✗ The system is broken into multiple components.
- ✗ Each component is assigned an appropriate delivery model (IaaS / PaaS / SaaS).
- ✗ At least **3-4 components** are analyzed.
- ✗ Each delivery model choice is technically justified.
- ✗ A shared responsibility matrix is included.

Quick self-check:

Did you avoid assigning a single delivery model to the entire system?

3. Deployment Model Decision

- ✗ A deployment model is clearly selected.
- ✗ Justification includes data sensitivity considerations.
- ✗ Governance and compliance factors are discussed.
- ✗ Cost implications are explained.
- ✗ Scalability requirements are considered.

Quick self-check:

Did you explain *why this model is appropriate*, not just what it is?

4. Architecture Diagram Quality

- ✗ Diagram clearly shows system layers.
- ✗ Major components are labeled.
- ✗ Delivery model for components is indicated.
- ✗ Deployment boundaries are shown.
- ✗ Trust boundaries are illustrated.
- ✗ Security components are included.

Quick self-check:

If someone sees only the diagram, can they understand how the system works?

5. Cloud Characteristics Implementation

- ✗ Each cloud characteristic is linked to a specific part of the architecture.
- ☐ Explanation focuses on implementation, not definition.
- ☐ Operational impact is discussed.
- ✗ Architecture supports scalability and reliability.

Quick self-check:

Did you explain *how the architecture enables the characteristic*?

6. Risk and Governance Analysis

- ✗ At least **five risks** are identified.
- ✗ Each risk has a root cause explanation.
- ☐ Each risk includes architectural impact.
- ✗ Each risk includes a mitigation strategy.
- ✗ Responsibility (provider vs consumer) is identified.

Quick self-check:

Are your risks specific to your architecture?

7. Engineering Reasoning

- ✗ Architectural decisions are logically justified.
- ✗ Trade-offs are acknowledged.
- ✗ Cost vs control considerations are discussed.
- ✗ Limitations of the design are recognized.

Quick self-check:

Did you explain **why your architecture makes sense**, not just what it is?

8. Academic Integrity

- ✗ All explanations are written in your own words.
- ✗ No diagrams are copied directly from the internet.
- ✗ External references (if used) are properly cited.

9. Formatting Compliance

- ✗ Report length is **7–8 pages maximum**.
- ✗ Sections follow the provided template.
- ✗ Font and spacing follow assignment instructions.
- ✗ Tables and diagrams are readable.

10. Final Engineering Check

Answer these questions in Yes/No/Not Clear:

- Can this architecture **scale if user traffic increases**?
Yes · Auto-scaling groups and load balancers handle traffic spikes
- Can it **handle failures without major disruption**?

Yes · Multi-zone and multi-region deployment ensure availability

- Does it **protect sensitive data properly**?

Yes · Payment/user data isolated in private cloud; encryption + IAM applied

- Does it **balance cost and control effectively**?

Yes · Public cloud reduces cost; private cloud maintains control over sensitive data

If the answer to these questions is **no/not clear**, improve your design.

Remarks: *(If there is anything different that you would like to explain about your assignment, explain here)*

Cloud characteristics section could be improved with deeper operational impact discussion. Risk table Impact column describes business
